Khazar Ahmadi

khazar.ahmadi@rub.de | Google Scholar | Orcid
https://khazarahmadi.github.io/

Research Overview

My research centers on uncovering the neural underpinnings of visual perception and high-level cognitive functions across the spectrum of health, aging, and disease. To achieve this, I employ a multidisciplinary approach that integrates state-of-the-art neuroimaging, behavioral paradigms, and computational modeling to unravel the fine-grained functional and structural architecture of the human brain, with a particular emphasis on the medial temporal lobe regions.

Academic Experience

Ruhr University Bochum — *Bochum, Germany*
Postdoctoral Researcher with Prof. Nikolai Axmacher — 2022 - Present

National Institutes of Health (NIH) — *Bethesda, USA*
Visiting Researcher with Dr. Renzo Huber — 2024

Lund University — *Lund, Sweden*
Postdoctoral Researcher with Prof. Oskar Hansson — 2020 - 2022

Spinoza Center for Neuroimaging — *Amsterdam, Netherlands*
Visiting Researcher with Prof. Serge Dumoulin — 2017

University Medical Center Groningen — *Groningen, Netherlands*
Visiting Researcher with Prof. Frans Cornelissen — 2017

Education

Otto-von-Guericke University — *Magdeburg, Germany*
PhD in psychology (magna cum laude) — 2014 - 2019

- Thesis title: Plasticity and stability of the cortical wiring in the human visual system
- Advisor: Prof. Michael Hoffmann

Shahid Beheshti University — *Tehran, Iran*
M.Sc. in child and adolescent clinical psychology (with distinction) — 2010 - 2013

Tabriz University — *Tabriz, Iran*
B.Sc. in clinical psychology (with distinction) — 2006 - 2010

Publications

Submitted Manuscripts

- Ahmadi, K., Swegle, S., Kashyap, S., Bouyeure, A., Bandettini, P., Axmacher, N., Huber, R. (2025). Blood volume sensitive laminar fMRI with VASO in human hippocampus: Capabilities and biophysical challenges at clinical 7T scanners. bioRxiv. [pre-print link]

- Rau, E.M.B., Heinen, R., Herweg, N.A., Patai, E.Z., Kobelt, M., **Ahmadi, K.**, Axmacher, N. (2025). Task-relevant cognitive maps in episodic memory. bioRxiv. [pre-print link]

Published Manuscripts

- Anijärv, T., Ossenkoppele, R., Smith, R., Binette, A.P., Collij, L., Behjat, H., Rittmo, J., Karlsson, L., **Ahmadi, K.**, Strandberg, O., van Westen, D., Vogel, J., Stomrud, E., Palmqvist, S., Mattsson-Carlgren, N., Spotorno, N., Hansson, O. (2025). Hemispheric asymmetry of tau pathology is related to asymmetric amyloid deposition in Alzheimer's disease. Nature Communications. [Link]

- **Ahmadi, K.**, Pereira, J., van Westen, D., Pasternak, O., Zhang, F., Nilsson, Stomrud, E., Spotorno, N. and Hansson, O. (2024). Fixel-based analysis reveals tau-related white matter changes in early stages of Alzheimer's disease. Journal of Neuroscience. [Link]

- Molz, B., Herbik, A., Baseler, H., de Best, P., RAZ, N., Gouws, A., **Ahmadi, K.**, Lowndes, R., McLean, R., Gottlob, I., Kohl, S., Choritz, L., Maguire, J., Kanowski, M., Käsmann-Kellner, B., Wieland, I., Banin, E., Levin, N., Morland, A., and Hoffmann, M.B. (2023). Achromatopsia - limits to visual cortex plasticity in the absence of functional cones. Investigative Ophthalmology & Visual Science [Link]

- Cicognola, C., Mattsson-Carlgren, N., van Westen, D., Zetterberg, H., Blennow, K., Palmqvist, S., **Ahmadi, K.**, Strandberg, O., Stomrud, E., Janelidze, S., Hansson, O. (2023). Associations of CSF PDGFRβ with aging, blood-brain barrier damage, neuroinflammation and Alzheimer disease pathological changes. Neurology. [Link]

- **Ahmadi, K.**, Pereira, J. B., Berron, D., Vogel, J., Ingala, S., Strandberg, O. T., Janelidze, S., Barkhof, F., Pfeuffer, J., Knutsson, L., van Westen, D., Palmqvist, S., Mutsaerts, H. J., Hansson, O. (2022). Gray matter hypoperfusion is a late pathological event in the course of Alzheimer's disease. Journal of Cerebral Blood Flow & Metabolism. [Link]

- Molz, B., Herbik, A., Baseler, H. A., de Best, P., Vernon, R., Raz, N., Gouws, A. D., **Ahmadi, K.**, Lowndes, R., Mclean, R. J., Gottlob, I., Kohl, S., Choritz, L., Maguire, J., Kanowski, M., Käsmann-Kellner, B., Wieland, I., Banin, E., Levin, N., Hoffmann, M. B., Morland, A. B. (2021). Structural changes to primary visual cortex in the congenital absence of cone input in achromatopsia. NeuroImage: Clinical. [Link]

- Puzniak, R. J., McPherson, B.*, **Ahmadi, K.***, Herbik, A., Kaufmann, J., Liebe, T., Gouws, A. D., Morland, A. B., Gottlob, I., Hoffmann, M. B., Pestilli, F. (2021). CHIASM, the human brain albinism and achiasma MRI dataset. Scientific Data. [* shared second-authorship; Link]

- Lowndes, R., Molz, B., Warriner, L., Herbik, A., de Best, P., Raz, N., Gouws, A. D., **Ahmadi, K.**, Mclean, R. J., Gottlob, I., Kohl, S., Choritz, L., Maguire, J., Kanowski, M., Käsmann-Kellner, B., Wieland, I., Banin, E., Levin, N., Hoffmann, M. B., Morland, A. B., Baseler, H. (2021). Structural differences across multiple visual cortical regions in the absence of cone function in congenital achromatopsia. Frontiers in Neuroscience. [Link]

- **Ahmadi, K.**, Fracasso, A., Puzniak, R. J., Gouws, A. D., Yakupov, R., Speck, O., Kaufmann, J., Pestilli, F., Dumoulin, S. O., Morland, A. B., Hoffmann, M. B. (2020).Triple visual hemifield maps in optic chiasm hypoplasia. NeuroImage. [Link]

- Carvalho, J., Invernizzi, A., **Ahmadi, K.**, Hoffmann, M. B., Renken, R. J., Cornelissen, F. W. (2020).Micro-probing enables fine-grained mapping of neuronal populations using fMRI. NeuroImage. [Link]

- Puzniak, R. J., **Ahmadi, K.**, Kaufmann, J., Gouws, A. D., Morland, A. B., Pestilli, F., Hoffmann, M. B. (2019). Quantifying nerve decussation abnormalities in the optic chiasm. NeuroImage: Clinical. [Link]

- **Ahmadi, K.**, Herbik, A., Wagner, M., Kanowski, M., Thieme, H., Hoffmann, M. B. (2019). Population receptive field and connectivity properties of the early visual cortex in human albinism. NeuroImage. [Link]

- Eick, C., **Ahmadi, K.**, Sweeney-Reed, C. M., Hoffmann, M. B. (2019). Interocular transfer of visual memory: influence of visual impairment and abnormalities of the optic chiasm. Neuropsychologia. [Link]

- **Ahmadi, K.**, Fracasso, A., van Dijk, J. A., Kruijt, C., van Genderen, M., Dumoulin, S. O., & Hoffmann, M. B. (2018). Altered organization of the visual cortex in FHONDA syndrome. NeuroImage. [Link]

- Arngrim+, N., Hougaard+, A., **Ahmadi, K.***, Vestergaard*, M. B., Schytz, H. W., Amin, F. M., Larsson, H.B.W., Olesen, J., Hoffmann, M.B. & Ashina, M. (2017). Heterogenous migraine aura symptoms correlate with visual cortex functional magnetic resonance imaging responses. Annals of Neurology. [+ & * shared first & second-authorship; Link]

- Hoffmann, M. B., Thieme, H., & **Ahmadi, K.** (2017). Potential of fMRI for the Functional Assessment of the Pathological Visual System. Klinische Monatsblätter für Augenheilkunde. [Link]

- **Ahmadi, K.**, Pouretemad, H. R., Esfandiari, J., Yoonessi, A., Yoonessi, A. (2015). Psychophysical evidence for impaired magno, parvo, and konio-cellular pathways in dyslexic children. Journal of Ophthalmic & Vision Research. [Link]

Conference Presentations

Talks

- **Ahmadi, K.** (2025). BOLD and non-BOLD contrasts for laminar fMRI at ultra-high field in human hippocampus. Spring Hippocampal Research Conference, Verona, Italy.
- **Ahmadi, K.** (2024). Laminar profile of hippocampal subregions during spatial navigation. International Society for Magnetic Resonance in Medicine (ISMRM), Singapore, Singapore.
- **Ahmadi, K.** (2022). Tau accumulation is associated with fiber-specific white matter degeneration in Alzheimer's disease. International Society for Magnetic Resonance in Medicine (ISMRM), London, England.
- **Ahmadi, K.** (2022). Disentangling fiber-specific white matter alterations in relation to hallmarks of Alzheimer's disease. Alzheimer's Disease and Parkinson's Disease (ADPD), Barcelona, Spain.
- **Ahmadi, K.** (2021). Altered cerebral blood perfusion in Alzheimer's disease spectrum and its association with amyloid-β and tau pathology. Alzheimer's Disease and Parkinson's Disease (ADPD), Virtual conference.
- **Ahmadi, K.** (2018). Macroscopic and mesoscopic cortical organization in congenital visual pathway abnormalities. Brain-in-depth (BID) - Symposium on layer-dependent MRI, German Center for Neurodegenerative Diseases (DZNE), Magdeburg, Germany.
- **Ahmadi, K.** (2016). Altered retino-cortical connections and visual cortex reorganization in the recently discovered FHONDA syndrome. Society for Neuroscience (SfN), San Diego, USA.

Poster Presentations

- **Ahmadi, K.**, Swegle, S., Rubin, M., Morgan, A. T., Bouyeure, A., Bandettini, P., Axmacher, N., Huber R. (2025). Blood volume sensitive laminar fMRI with VASO in human hippocampus: Capabilities of clinical 7T and biophysical challenges. International Society for Magnetic Resonance in Medicine (ISMRM), Hawaii, USA.
- **Ahmadi, K.**, Stawarczyk, D., Pfaffenrot, V., Gomes, C. A., Kashyap, S., Patai, Z., Norris, D. G., Axmacher, N. (2024). Laminar profiles of hippocampal subfields are differentially associated with navigation strategies. Organization for Human Brain Mapping (OHBM), Seoul, south Korea.
- Rau, E. M. B., Herweg, N. A., Heinen, R., **Ahmadi, K.**, Axmacher, N. (2024). Functions of the medial temporal lobe in memory and navigation of abstract conceptual spaces. Federation of European Neuroscience Societies (FENS), Vienna, Austria.
- Kashyap, S., Gomes, C. A., **Ahmadi, K.**, Axmacher, N., Uludag, K. (2024). A new proposed workflow for processing high spatial resolution MRI data with multiple sessions. Cognitive Computational Neuroscience (CCN), Boston, USA.
- **Ahmadi, K.**, Pereira, J. B., Berron, D., Vogel, J., Ingala, S., Strandberg, O. T., Janelidze, S., Barkhof, F., Pfeuffer, J., Knutsson, L., van Westen, D., Mutsaerts, H. J., Palmqvist, S., Hansson, O. (2021). Tau and synaptic biomarkers but not amyloid-β are associated with cerebral perfusion in the Alzheimer's disease spectrum. Alzheimer's Association International Conference (AAIC), Virtual conference.
- Cicognola, C., Janelidze, S., Van Westen, D., **Ahmadi, K.**, Hansson, O. (2021). Cerebrospinal fluid platelet-derived growth factor receptor beta measured in BioFinder-2: an early biomarker in the Alzheimer's disease continuum? Clinical Trials on Alzheimer's Disease (CTAD), Boston, USA.
- Puzniak, R. J., McPherson, B., **Ahmadi, K.**, Herbik, A., Kaufmann, J., Liebe, T., Gouws, A. D., Morland, A. B., Gottlob, I., Hoffmann, M. B., Pestilli, F. (2020). Chiasmal malformations dataset: a unique neuroimaging testbed. Vision Science Society (VSS), Virtual conference.
- Hoffmann, M. B., Molz, B., Herbik, A., de Best, P., Raz, N., Gouws, A. D., **Ahmadi, K.**, Lowndes, R., McLean, R. J., Kohl, S., Gottlob, I., Choritz, L., Maguire, J., Kanowski, M., Käsmann-Kellner, B., Wieland, I., Banin, E., Levin, N., Baseler, H. Morland, A. B. (2020).

Visual cortex stability and plasticity in the absence of functional cones in achromatopsia. Vision Science Society (VSS), Virtual conference.

- **Ahmadi, K.**, Fracasso, A., Puzniak, R. J., Gouws, A. D., Yakupov, R., Speck, O., Kaufmann, J., Pestilli, F., Dumoulin, S. O., Morland, A. B., Hoffmann, M. B. (2019). Triple hemi-field input to the visual cortex in a patient with chiasmal hypoplasia. Organization for Human Brain Mapping (OHBM), Rome, Italy.
- Carvalho, J., **Ahmadi, K.**, Invernizzi, A., Hoffmann, M. B., Renken, R., Cornelissen, F. W. (2019). Micro-probing the visual cortex: high-resolution mapping of neuronal sub-populations. Organization for Human Brain Mapping (OHBM), Rome, Italy.
- Puzniak, R. J., **Ahmadi, K.**, Kaufmann, J., Gouws, A. D., Morland, A. B., Pestilli, F., Hoffmann, M. B. (2019). Quantification of nerve decussation abnormalities in the optic chiasm. Organization for Human Brain Mapping (OHBM), Rome, Italy.
- Prabhakaran, G., Al-Nosairy, K. O., Tempelmann, C., **Ahmadi, K.**, Hoffmann, M. B. (2019). Brain activity in input-deprived visual cortex in glaucoma – no fMRI-evidence of plasticity. Organization for Human Brain Mapping (OHBM), Rome, Italy.
- **Ahmadi, K.**, Fracasso, A., Gouws, A. D., Morland, A. B., Dumoulin, S. O., Hoffmann, M. B. (2017). Functional organization of the visual cortex in a unique case of achiasma. Society for Neuroscience (SfN), Washington, D.C., USA.
- **Ahmadi, K.**, Fracasso, A., van Dijk, J. A., Kruijt, C., van Genderen, M., Dumoulin, S. O., & Hoffmann, M. B. (2017). Cortical plasticity in FHONDA: a new inherited visual system disorder. European conference on visual perception (ECVP), Berlin, Germany.
- **Ahmadi, K.**, Pouretemad, H. R., Esfandiari, J., Yoonessi, A. (2013). Investigation of magno, parvo and koniocellular pathways in children with dyslexia. International Conference of Cognitive Sciences (ICCS), Tehran, Iran.

Technical Skills

Neuroimaging Modalities	fMRI (at 7 and 3 T), dMRI, sMRI & ASL
Neuroimaging Software	Nipype, AFNI, FSL, MRtrix, DIPY, FreeSurfer, LayNii, ExploreASL, HippUnfold, ASHS, SPM, ANTs & MrVista
Neuroimaging Hardware	Siemens 7T MAGNETOM Terra Scanner Operator
Programming	Bash, R, Python, MATLAB, LaTeX
Experiment Software	PsychoPy, Psychtoolbox, Neurobehavioral Systems Presentation
Languages	Azerbaijani (native), Persian (native), English (fluent), German (upper-intermediate fluency; Level B2)

Awards and Honors

Finalist, ISMRM trainee poster award *2025*
Selected among the top five abstracts for the poster award competition at the ISMRM High Field Study Group
Hawaii, USA

Winner, Erwin L. Hahn Institute's brain-art competition *2023*
Awarded € 200 for the art-work "Hippocampal Moon"
Essen, Germany

Greta & Johan Kock Foundation grant for medical research *2021*
(About)
Lund, Sweden
Awarded € 10,000 for the project "Clinical Use of Novel dMRI Markers in Alzheimer's Disease"

Lund University MultiPark travel grant *2021-2022*
(About) Lund, Sweden
Awarded a total of € 2,000, covering conference registrations and travel expenses

The ISMRM Summa Cum Laude Merit Award *2022*
(About) London, UK
- Selected abstract in top 5% of all submissions

Marie Curie Ph.D Fellowship *2015-2018*
Research Training in computational Visual Neuroscience Magdeburg, Germany

Top Student Award *2010*
Ranked 1st out of 50 students based on cumulative GPA in BSc program,
received a cash prize of € 100 Tabriz, Iran

Teaching & Mentorship

Ruhr University Bochum *Bochum, Germany*
Lead instructor, "fMRI: from Theory to Practice" 2022 - Present
Lead instructor, "Aging, dementia and memory disorders" 2024
Mentored students:
- Dana Schröder & Paul Höchter (Research assistants)
- Kübra Karatas & Niko Britt (MSc. & BSc. Interns)
- Alena Machentanz (BSc. thesis title: "Memory performance in active vs. passive learning contexts")

Co-mentored students:
- Elias Rau & Javier Schneider-Penate (PhD students)

Neuromatch Academy: Computational Neuroscience Workshop *Online Event*
Mentor of the group "NeuroMatched Philosophers" 2021
- Project: predicting fearful stimuli from HCP fMRI data using MVPA

Otto-von-Guericke University *Magdeburg, Germany*
Instructor: "Neuroscience Journal Club" 2016 - 2018
Co-mentored students:
- Charlotta Eick (MSc. thesis title: "Inter-ocular transfer of visual memory in albinism")
- Gokulraj Prabhakaran (PhD student) & Anne Herbik (Postdoc with no neuroimaging experience

Professional Service

Ad-hoc Reviewer
NeuroImage, Cell Reports Medicine, Alzheimer's & Dementia, Alzheimer's Research & Therapy, Journal of Cerebral Blood Flow & Metabolism, Brain Communications, Scientific Reports, Neuropsychologia, Brain Research Bulletin, British Journal of Ophthalmology

Symposia & Lecture Organization *Bochum, Germany*
- Advances in Layer-specific fMRI
- Surface-based analysis for hippocampal subfield segmentation

2023

Public Outreach *Magdeburg, Germany*
- Long Night of Science at Otto-von-Guericke University 2016 - 2019

Oslo, Norway
- Popular-science talk at 4th European Days of Albinism 2018